

Module 3 Notes — Extract and Load (EL): landing files in BigQuery

Your notes for the first after-lunch module on Day 1 (Mon Oct 5, 2026, 13:15–13:45 CT). It sets up the hands-on [loading block](#) that follows at 13:45, where you'll do all of this yourself.

Key takeaways

- **Extract & Load is your Bronze-landing pattern for files.** Datastream (this morning) covers databases that never stop changing; EL covers the other half of your world — files that arrive on a schedule. Land the file in BigQuery as-is, transform later. That "land first, transform later" shape is exactly your Bronze layer.
- **One load job, three front doors.** The console's Create-table form, the bq load command, and the LOAD DATA SQL statement all submit the same kind of load job. Pick per situation: console for exploring, CLI for scripts, SQL for anything that must live in a procedure or a scheduled query.
- **Batch loads are free.** Loads run on a shared pool of load capacity, not on your query slots. Queries bill by bytes scanned; loads don't bill at all. (You'll verify "Bytes billed: 0 B" with your own eyes in the loading block.)
- **Loads append by default.** Re-running a load adds the rows again; a replace/overwrite flag truncates first. Decide deliberately which one each pipeline means.
- **Autodetect is a starting point, not a pipeline.** It samples the start of a file and guesses types. Fine for a first look; never for a recurring feed. Pin schemas for anything that runs more than once — the loading block shows you the failure that taught this rule.
- **File formats decide load speed.** A single gzipped CSV can't be split across workers; Parquet is splittable, columnar, binary, and carries its own schema. You'll race the same 60 million rows both ways — minutes versus seconds.
- **BigQuery Data Transfer Service is scheduled ingest as a service.** It automates recurring loads — from SaaS applications, Cloud Storage, and more — on a schedule. Think of it as the managed answer to the SQL Agent import job.
- **External tables let you query data where it lives.** You define a table in BigQuery over files that stay in Cloud Storage (or Google Sheets); there is no load step and the data doesn't move. Good for exploration and for data you rarely scan.
- **BigLake tables are external tables with warehouse manners.** They add governance and performance features — fine-grained security, metadata caching — over files in open formats. This is the bridge to Day 2's lakehouse and your Bronze → Silver → Gold design.
- **The table-type choice is per dataset, not per platform.** Managed, external, and BigLake tables coexist in the same project and the same queries. Your medallion layers can mix them deliberately.

Your SQL Server world → GCP world

Your SQL Server world today	The GCP world you're moving to
BULK INSERT / OPENROWSET(BULK)	bq load, or LOAD DATA ... FROM FILES in SQL
Import and Export wizard	Console: Create table from Google Cloud Storage
SSIS flat-file import package	A scripted bq load, or LOAD DATA inside a scheduled query
SQL Agent job importing a daily file	BigQuery Data Transfer Service (or a scheduled query)
Staging file share (\\server\drops)	A Cloud Storage bucket — your Bronze landing zone
Linked server pointing at external files	External table over a Cloud Storage bucket
Fixed staging-table schema in source control	A pinned schema JSON file in source control

Managed vs external vs BigLake (lakehouse) tables

	Managed table	External table	BigLake table
Where the bytes live	BigQuery- managed storage	Your files (Cloud Storage, Sheets, ...)	Your files in Cloud Storage, in open formats
Where the schema/types come from	BigQuery	Declared in BigQuery, read from the files at query time	Declared in BigQuery; formats like Parquet carry real types
Query performance	Full BigQuery storage optimizations	Files read at query time	External, plus performance features such as metadata caching
Governance	Full BigQuery security	Limited	Fine-grained (row/column) security over the files
Likely fit in your migration	Silver/Gold data marts	Exploration; rarely-scanned data	The Bronze lakehouse zone — Day 2's subject

Dig deeper

- [Loading data into BigQuery](#)
- [The bq command-line tool](#)

- [LOAD DATA statement reference](#)
- [BigQuery Data Transfer Service](#)
- [Querying external data sources](#)
- [BigLake tables](#)

Part of the Iowa DOM DoIT Data Engineering student pack — see ../00_schedule.md.